

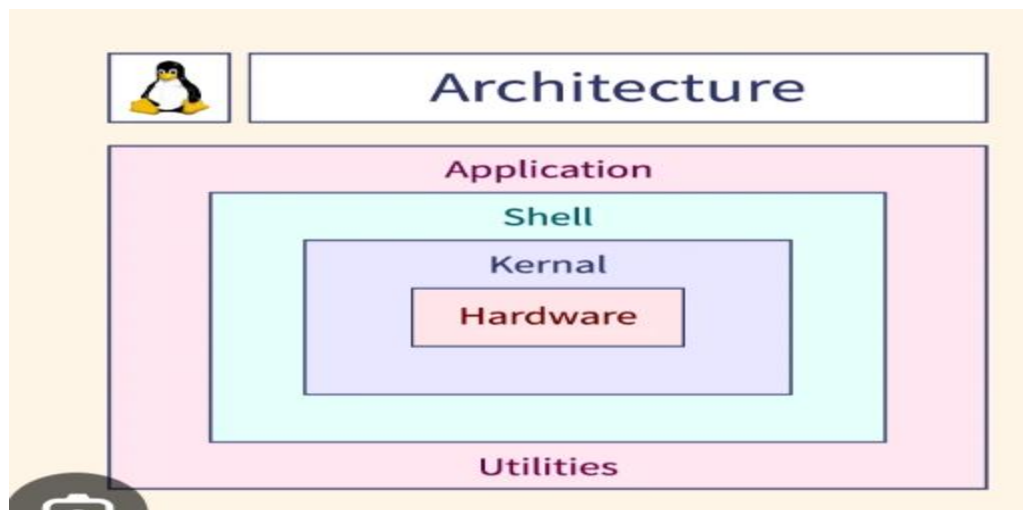
Day –1 Introduction to DevOps and cloud:

Task: Today's goal is to set foundation for your DevOps Journey

1. *Mention your current level: Student*
2. *Define 3 clear goals for the next 90 days:*
 1. Understand DevOps concepts clearly and build projects independently.
 2. Work on real-time, hands-on DevOps projects.
 3. To get a job.
3. *Define 3 Core DevOps skills:*
 1. Linux troubleshooting and system fundamentals.
 2. CI/CD pipeline implementation and automation.
 3. Kubernetes deployment and debugging.
4. *Weekly Time Commitment:*
 - Weekdays: 3-3.5 hours per day.
 - Weekends: 5 hours per day.

Day-2: Linux Architecture, Processes, and systemd:

Today's goal is to understand how linux works under the hood.



You will create a short note that explains: The core components of Linux (kernel, user, systemd), How processes are created and managed, what systemd does and why it matters.

1. Core components:

- Kernel: The kernel is the core of the os. It runs in kernel space and has full hardware access.
- User Space: User Space is where applications run on Nginx and chrome or shell or Utilities (ls,ps,grep). Or Environment where users and applications operate.
- Init System: After the kernel boots, it starts PID 1, which is the init system.

2. How Processes are created and Managed:

- What is a process: A process is a running instance of a program. Every running program is a process.
- Common Process states: R = Running (actively using CPU), S = Sleeping (waiting resource), T = Stopped (Manually paused), Z = Zombie (finished but not cleaned up).

3. Systemd : Service Management = is the modern init system used in most linux distributions.

- Responsibilities: Starts service during system boot. Automatically restarts failed services, Manages service dependencies.

Day –3: Linux Commands Practice:

Process Management Commands:

1. Ps aux = shows all running processes with detailed info.
2. Top = Shows CPU monitor.
3. htop = Interactive process viewer.
4. Free -h = show memory usage in human-readable format.
5. Uptime = show system uptime and load average.
6. pstree = shows running processes as a tree diagram to display parent-child relationships.
7. Kill = terminate the process using PID.
8. jobs : Lists the status of jobs started in the current terminal session.

File system:

1. pwd = prints the present working directory.
2. ls = list all files and directory.

3. cd = change the current directory to a new specified path.
4. find = searches for files or directory based on names.
 4. mkdir = create one or more new directories.
 5. Touch = creates a new empty file.
 6. rmdir = remove empty directories.
 7. Cat = display the entire content of a file.
 8. Head = displays the first 10 lines of a file.
 9. Tail = Displays the last 10 lines of a file.
 10. Grep = searches for specific text patterns within files.
 11. df -h = displays disk usage.
 12. mount = Attaches a separate file system.
 13. umount = safely detaches a mounted file system.
 14. Chmod = changes the read, write and execute permissions of a file or directory.
 15. Chown = changes the user and /or group ownership of a file.
 16. Ls -l: displays the lists of files and displays.

Networking commands:

1. ping: sends small packets of data to a network.
2. ip addr : Displays ip addresses and network interface.
3. **netstat**: Shows active network connections, routing tables, and interface statistics (often replaced by the faster **ss** command).
4. **hostname**: Shows or sets the system's network name.



Day 4/9 - Practice Linux fundamentals with real commands.

1. processes commands: (ps, top, htop, pgrep)
2. Service commands: (systemctl status, systemctl list-units, systemctl enable restart)
3. Log commands: (journalctl -u <service>, tail -n 100)
4. While inspecting the SSH service, I realized that restarting a service doesn't just refresh it, it actually kills the old Process ID (PID) and spawns a new one. Watching the PID change in real time made me understand what restart truly means at the process level.
you can also explore this by entering : (command : systemctl status ssh now you can look for this line - Main PID: 1234 (sshd))

👉 Note down that PID number.

now lets Restart the service : `sudo systemctl restart ssh` , now this command stops the service, kills the current process , starts a new process.

now let's check the PID again : use this command (`systemctl status ssh`) now you can see Main PID: 1458 (sshd). The PID will be different.

Day 5: Linux Troubleshooting Drill CPU, Memory, and Logs.

1. Choose Target Service: `systemctl status ssh`.

2. **Environment basics** (Kernel info: `uname -r`, OS version: `cat /etc/os-release`)

```
sri@LAPTOP-KH3CPCFO:~$ uname -a
Linux LAPTOP-KH3CPCFO 5.15.167.4-microsoft-standard-WSL2 #1 SMP T
ue Nov 5 00:21:55 UTC 2024 x86_64 x86_64 x86_64 GNU/Linux
sri@LAPTOP-KH3CPCFO:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.4 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-polic
ies/privacy-policy"
UBUNTU_CODENAME=jammy
```

3. Filesystem sanity: (create Temporary Folder - `mkdir /tmp/runbook-demo`, Copy System file - `cp /etc/hosts /tmp/runbook-demo/hosts-copy`, to verify: `ls -l /tmp/runbook-demo`)

```
sri@LAPTOP-KH3CPCFO:~$ mkdir /tmp/runbook-demo
sri@LAPTOP-KH3CPCFO:~$ cp /etc/hosts /tmp/runbook-demo/hosts-copy
sri@LAPTOP-KH3CPCFO:~$ ls -l /tmp/runbook-demo
total 4
-rw-r--r-- 1 sri sri 416 Feb 13 00:43 hosts-copy
sri@LAPTOP-KH3CPCFO:~$ |
```

4. CPU memory snapshot: (`grep sshd` , `top` , `htop` , `free -h`)

```
sri@LAPTOP-KH3CPCFO: ~  
sri@LAPTOP-KH3CPCFO:~$ pgrep sshd  
4350  
sri@LAPTOP-KH3CPCFO:~$ ps -o pid  
error: garbage option  
  
Usage:  
ps [options]  
  
Try 'ps --help <simple|list|output|threads|misc|all>'  
or 'ps --help <s|l|o|t|m|a>'  
for additional help text.  
  
For more details see ps(1).  
sri@LAPTOP-KH3CPCFO:~$ free -h  
  
Usage:  
free [options]  
  
Options:  
-b, --bytes          show output in bytes  
--kilo               show output in kilobytes  
--mega              show output in megabytes  
--giga              show output in gigabytes  
--tera             show output in terabytes  
--peta             show output in petabytes  
-k, --kibi          show output in kibibytes  
-m, --mebi          show output in mebibytes  
-g, --gibi          show output in gibibytes  
--tebi             show output in tebibytes  
--pebi             show output in pebibytes  
-h, --human         show human-readable output
```

```
sri@LAPTOP-KH3CPCFO:~$ free -h  
  
Usage:  
free [options]  
  
Options:  
-b, --bytes          show output in bytes  
--kilo               show output in kilobytes  
--mega              show output in megabytes  
--giga              show output in gigabytes  
--tera             show output in terabytes  
--peta             show output in petabytes  
-k, --kibi          show output in kibibytes  
-m, --mebi          show output in mebibytes  
-g, --gibi          show output in gibibytes  
--tebi             show output in tebibytes  
--pebi             show output in pebibytes  
-h, --human         show human-readable output  
--si                use powers of 1000 not 1024  
-l, --lohi          show detailed low and high memory statistics  
-t, --total         show total for RAM + swap  
-s N, --seconds N   repeat printing every N seconds
```

5. Disk I/O: (dh -h, du -sh, vmstat):


```

sri@LAPTOP-KH3CPCF0:~$ vmstat
procs-----memory-----swap--io-----system--cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st
0 0 0 5835300 15572 452752 0 0 27 15 11 65 0 0 99 0 0
sri@LAPTOP-KH3CPCF0:~$ dstat
Command 'dstat' not found, but can be installed with:
sudo apt install dstat # version 0.7.4-6.1, or
sudo apt install pcps # version 5.3.6-1build1
sri@LAPTOP-KH3CPCF0:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G  0 3.8G  0% /usr/lib/modules/5.15.167.4-microsoft-standard-WSL2
none            3.8G  4.0K 3.8G  1% /mnt/wsl
drivers          476G  238G  239G  50% /usr/lib/wsl/drivers
/dev/sdc         1007G   11G  945G   2% /
none            3.8G   92K 3.8G  1% /mnt/wslg
none            3.8G  0 3.8G  0% /usr/lib/wsl/lib
rootfs           3.8G  2.2M 3.8G  1% /init
none            3.8G  940K 3.8G  1% /run
none            3.8G  0 3.8G  0% /run/lock
none            3.8G  0 3.8G  0% /run/shm
tmpfs            4.0M  0 4.0M  0% /sys/fs/cgroup
none            3.8G   96K 3.8G  1% /mnt/wslg/versions.txt
none            3.8G   96K 3.8G  1% /mnt/wslg/doc
C:\              476G  238G  239G  50% /mnt/c
tmpfs            3.8G  0 3.8G  0% /run/qemu
snapfuse         128K  128K  0 100% /snap/bare/5
snapfuse         74M   74M  0 100% /snap/core22/2216
snapfuse         74M   74M  0 100% /snap/core22/2292
snapfuse         67M   67M  0 100% /snap/core24/1267
snapfuse         67M   67M  0 100% /snap/core24/1349
snapfuse         92M   92M  0 100% /snap/gtk-common-themes/1535
snapfuse         50M   50M  0 100% /snap/snapd/24792
snapfuse         49M   49M  0 100% /snap/snapd/25935
snapfuse         32M   32M  0 100% /snap/terraform/837
snapfuse        132M  132M  0 100% /snap/ubuntu-desktop-installer/1276
snapfuse        132M  132M  0 100% /snap/ubuntu-desktop-installer/1286
sri@LAPTOP-KH3CPCF0:~$ du -sh
422M
sri@LAPTOP-KH3CPCF0:~$ vmstat
procs-----memory-----swap--io-----system--cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st
1 0 0 5888928 5096 407600 0 0 27 15 11 65 0 0 99 0 0
sri@LAPTOP-KH3CPCF0:~$

```

6. Network: (ss -tulpn , netstat -tulpn) :

```

sri@LAPTOP-KH3CPCF0:~$ ss -tulpn
Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process
udp    UNCONN 0        0       127.0.0.53%lo:53    0.0.0.0:*
udp    UNCONN 0        0       10.255.255.254:53   0.0.0.0:*
udp    UNCONN 0        0       0.0.0.0:111        0.0.0.0:*
udp    UNCONN 0        0       127.0.0.1:323      0.0.0.0:*
udp    UNCONN 0        0       0.0.0.0:49637      0.0.0.0:*
udp    UNCONN 0        0       0.0.0.0:33390      0.0.0.0:*
udp    UNCONN 0        0       127.0.0.1:908      0.0.0.0:*
udp    UNCONN 0        0       0.0.0.0:54594      0.0.0.0:*
udp    UNCONN 0        0       0.0.0.0:35118      0.0.0.0:*
udp    UNCONN 0        0       0.0.0.0:60431      0.0.0.0:*
udp    UNCONN 0        0       [::]:40956         [::]:*
udp    UNCONN 0        0       [::]:111           [::]:*
udp    UNCONN 0        0       [::1]:323          [::]:*
udp    UNCONN 0        0       [::]:50205         [::]:*
udp    UNCONN 0        0       [::]:34382         [::]:*
udp    UNCONN 0        0       [::]:34440         [::]:*
udp    UNCONN 0        0       [::]:39121         [::]:*
tcp    LISTEN 0        70      127.0.0.1:33060     0.0.0.0:*
tcp    LISTEN 0        1000    10.255.255.254:53  0.0.0.0:*
tcp    LISTEN 0        4096    0.0.0.0:39839      0.0.0.0:*
tcp    LISTEN 0        4096    0.0.0.0:51545      0.0.0.0:*
tcp    LISTEN 0        64      0.0.0.0:2049       0.0.0.0:*
tcp    LISTEN 0        4096    0.0.0.0:59023      0.0.0.0:*
tcp    LISTEN 0        151     127.0.0.1:3306     0.0.0.0:*
tcp    LISTEN 0        64      0.0.0.0:46837      0.0.0.0:*
tcp    LISTEN 0        4096    127.0.0.53%lo:53   0.0.0.0:*
tcp    LISTEN 0        4096    0.0.0.0:41017      0.0.0.0:*
tcp    LISTEN 0        128     0.0.0.0:22         0.0.0.0:*
tcp    LISTEN 0        4096    0.0.0.0:111        0.0.0.0:*
tcp    LISTEN 0        511     0.0.0.0:80         0.0.0.0:*

```

```
sri@LAPTOP-KH3CPCF0:~$ netstat -tulpn
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program
tcp        0      0 127.0.0.1:33060         0.0.0.0:*                LISTEN      -
tcp        0      0 10.255.255.254:53      0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:39839          0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:51545          0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:2049           0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:59023          0.0.0.0:*                LISTEN      -
tcp        0      0 127.0.0.1:3306         0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:46837          0.0.0.0:*                LISTEN      -
tcp        0      0 127.0.0.53:53          0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:41017          0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:22             0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:111            0.0.0.0:*                LISTEN      -
tcp        0      0 0.0.0.0:80             0.0.0.0:*                LISTEN      -
tcp6       0      0 :::44877               :::*                    LISTEN      -
tcp6       0      0 :::8080                :::*                    LISTEN      -
tcp6       0      0 :::45037               :::*                    LISTEN      -
tcp6       0      0 :::39409               :::*                    LISTEN      -
tcp6       0      0 :::39379               :::*                    LISTEN      -
tcp6       0      0 :::2049                :::*                    LISTEN      -
tcp6       0      0 :::42067               :::*                    LISTEN      -
tcp6       0      0 :::22                  :::*                    LISTEN      -
tcp6       0      0 :::111                 :::*                    LISTEN      -
tcp6       0      0 :::80                  :::*                    LISTEN      -
udp        0      0 127.0.0.53:53          0.0.0.0:*                -
udp        0      0 10.255.255.254:53      0.0.0.0:*                -
```

7. Logs : (journalctl -u ssh -n 50 , tail -n 50 /var/log/syslog)

```
sri@LAPTOP-KH3CPCF0:~$ journalctl -u ssh -n 50
Feb 13 00:17:02 LAPTOP-KH3CPCF0 systemd[1]: Starting OpenBSD Secure Shell s>
Feb 13 00:17:02 LAPTOP-KH3CPCF0 sshd[3681]: Server listening on 0.0.0.0 por>
Feb 13 00:17:02 LAPTOP-KH3CPCF0 sshd[3681]: Server listening on :: port 22.
Feb 13 00:17:02 LAPTOP-KH3CPCF0 systemd[1]: Started OpenBSD Secure Shell se>
Feb 13 00:19:10 LAPTOP-KH3CPCF0 sshd[3681]: Received signal 15; terminating.
Feb 13 00:19:10 LAPTOP-KH3CPCF0 systemd[1]: Stopping OpenBSD Secure Shell s>
Feb 13 00:19:10 LAPTOP-KH3CPCF0 systemd[1]: ssh.service: Deactivated succes>
Feb 13 00:19:10 LAPTOP-KH3CPCF0 systemd[1]: Stopped OpenBSD Secure Shell se>
Feb 13 00:19:10 LAPTOP-KH3CPCF0 systemd[1]: Starting OpenBSD Secure Shell s>
Feb 13 00:19:10 LAPTOP-KH3CPCF0 sshd[4350]: Server listening on 0.0.0.0 por>
Feb 13 00:19:10 LAPTOP-KH3CPCF0 sshd[4350]: Server listening on :: port 22.
Feb 13 00:19:10 LAPTOP-KH3CPCF0 systemd[1]: Started OpenBSD Secure Shell se>

sri@LAPTOP-KH3CPCF0:~$ tail -n 50 /var/log/syslog
Feb 13 00:17:00 LAPTOP-KH3CPCF0 systemd[1]: Reloading.
Feb 13 00:17:00 LAPTOP-KH3CPCF0 systemd[1]: Configuration file /run/systemd/
system/netplan-ovs-cleanup.service is marked world-inaccessible. This has no
effect as configuration data is accessible via APIs without restrictions. P
roceeding anyway.
Feb 13 00:17:01 LAPTOP-KH3CPCF0 systemd[1]: Reloading.
Feb 13 00:17:01 LAPTOP-KH3CPCF0 CRON[3621]: (root) CMD ( cd / && run-parts
--report /etc/cron.hourly)
Feb 13 00:17:01 LAPTOP-KH3CPCF0 systemd[1]: Configuration file /run/systemd/
system/netplan-ovs-cleanup.service is marked world-inaccessible. This has no
effect as configuration data is accessible via APIs without restrictions. P
roceeding anyway.
Feb 13 00:17:01 LAPTOP-KH3CPCF0 systemd[1]: Reloading.
Feb 13 00:17:01 LAPTOP-KH3CPCF0 systemd[1]: Configuration file /run/systemd/
system/netplan-ovs-cleanup.service is marked world-inaccessible. This has no
effect as configuration data is accessible via APIs without restrictions. P
roceeding anyway.
```

Quick Findings:

- Service runs normally.
- No CPU or memory spike
- Disk Usage healthy
- No log errors

If this worsens:

1. Restart SSH service safely
2. Increase SSH login verbosity in `/etc/ssh/sshd_config`
3. Run `strace -p <PID>` to trace system calls.
4. Check failed login attempts (`grep "Failed password"`)

Day 6 – Linux Fundamentals: Read and Write Text Files.

Today's goal is to practice basic file read/write using only fundamental commands.

1. create file : (`touch notes.txt`) verify now using this command : `ls -l notes.txt`
2. write text to a file :
 - `echo "Linux Day 06 Practice" > notes.txt`
 - `echo "Learning file operations" >> notes.txt`

note : `>` - overwrites, `>>` appends.

3. Append using `tee`(write and display at the same time) :
 - `echo "Practicing tee command" | tee -a notes.txt`

note : `-a`: append mode, without `-a` it overwrites.

4. Add more Lines:
 - `echo "Reading file using cat" >> notes.txt`
 - `echo "Reading first lines using head" >> notes.txt`
 - `echo "Reading last lines using tail" >> notes.txt`
5. Read full file: `cat notes.txt`
6. To read first 2 lines: `head -n 2 notes.txt`
7. To read last 2 lines: `tail -n 2 notes.txt`

Day 7– Linux File System Hierarchy & Scenario-Based Practice:

Part -1: Linux File System Hierarchy:

Core Directories:

/ (root): The starting point of everything.

/home: user home directory.

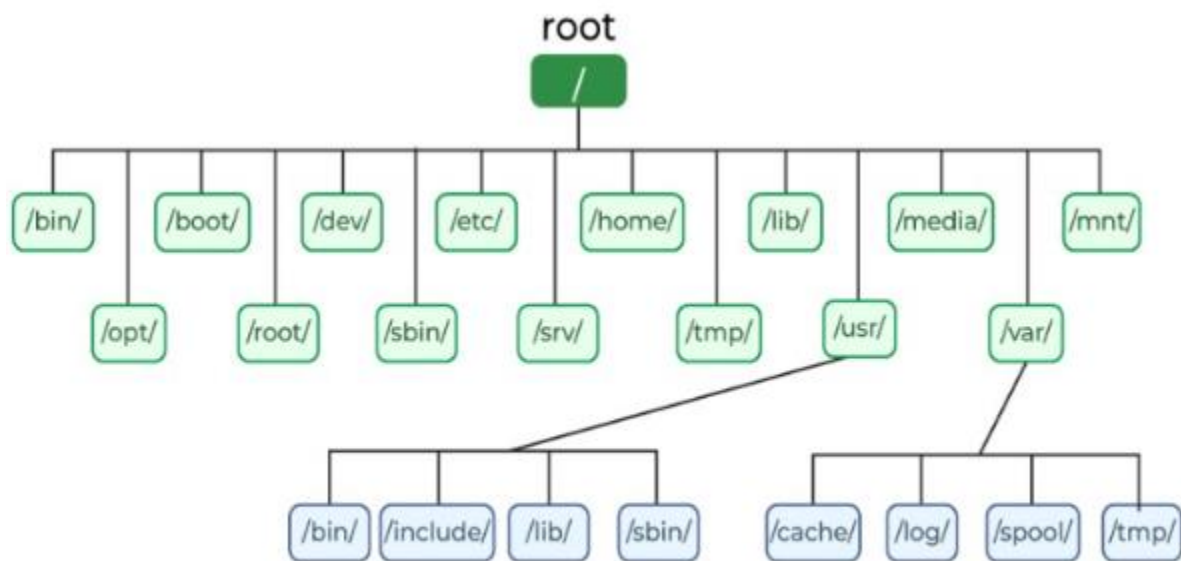
/etc: Configuration files.

/var/log: Log files.

/bin: Essential command binaries.

/usr/bin: user command binaries.

/opt: Optional/third-party applications.



Find the largest log file in /var/log: `du -sh /var/log/* 2>/dev/null | sort -h | tail -5`

Look at a config file in /etc: `cat /etc/hostname`

Check your home directory: `ls -la ~`

```

sri@LAPTOP-KH3CPCFO:~$ du -sh /var/log/* 2>/dev/null | sort -h | tail -5
832K    /var/log/kern.log
1.8M    /var/log/syslog.1
1.9M    /var/log/syslog
7.6M    /var/log/installer
801M    /var/log/journal
sri@LAPTOP-KH3CPCFO:~$ cat /etc/hostname
LAPTOP-KH3CPCFO
sri@LAPTOP-KH3CPCFO:~$ ls -la ~
total 56888
drwxr-x--- 31 sri sri      4096 Feb 14 15:44 .
drwxr-xr-x  5 root root     4096 Feb 12 18:49 ..
drwxr-xr-x  4 sri sri      4096 Jan 27 15:51 .ansible
lrwxrwxrwx  1 sri sri        31 Sep  1 2024 .aws -> /mnt/c/Users/VEERA VE
lrwxrwxrwx  1 sri sri        31 Sep  1 2024 .azure -> /mnt/c/Users/VEERA VE
lrwxrwxrwx  1 sri sri        31 Sep  1 2024 .awscli
-rw-r--r--  1 sri sri     23171 Feb 14 15:48 .bash_history
-rw-r--r--  1 sri sri       220 Aug  2 2024 .bash_logout
-rw-r--r--  1 sri sri     3771 Aug  2 2024 .bashrc
drwx----- 4 sri sri      4096 Jul  7 2025 .cache
drwx----- 4 sri sri      4096 Feb 11 23:56 .config
drwxr-xr-x  5 sri sri      4096 Sep  1 2024 .docker
drwxr-xr-x  3 sri sri      4096 Jul  7 2025 .dotnet
-rw-r--r--  1 sri sri    12288 Feb 17 2025 .firstfile.txt.swo
drwxr-xr-x  8 sri sri      4096 Sep  4 2024 .git
-rw-r--r--  1 sri sri       59 Aug 26 2024 .gitconfig
-rw-r--r--  1 sri sri       20 Feb 14 15:44 .lessht
drwxr-xr-x  4 sri sri      4096 Sep  3 2024 .local
-rw-r--r--  1 sri sri       0 Feb 14 15:32 .motd_shown
-rw-r--r--  1 sri sri       87 Aug 12 2024 .mysql_history
drwxr-xr-x  4 sri sri      4096 Jul  8 2025 .npm
-rw-r--r--  1 sri sri      807 Aug  2 2024 .profile
drwx----- 3 sri sri      4096 Jan 27 15:35 .ssh
-rw-r--r--  1 sri sri       0 Aug 12 2024 .sudo_as_admin_successful
drwxr-xr-x  2 sri sri      4096 Jan 31 18:28 .terraform.d
drwxr-xr-x  7 sri sri      4096 Feb 18 2025 .vagrant.d
-rw-r--r--  1 sri sri     6109 Jan 27 13:27 .viminfo
drwxr-xr-x  5 sri sri      4096 Jul  7 2025 .vscode-server
-rw-r--r--  1 sri sri       234 Jan 31 17:59 .wget-hsts
-rw-r--r--  1 sri sri     1933 Feb 13 00:12 .wsl-config

```

Part 2: Scenario-Based Practice.

Solved scenario: understanding how to approach scenarios.

Example scenario: Check if a service is running.

Question: How do you check if the nginx service is running?

1. Check service status: `systemctl status nginx`
Why this command? To see what service is active, failed, or stopped.
2. If service is not found, list all services: `systemctl list-units --type=service`
Why this command? To see what service exists on the system.
3. check if service is enabled on boot: `systemctl is-enabled nginx`
Why this command? To know if it will start automatically after reboot.

What I learned: Always check status first, then investigate based on what you see.

Scenario 1: Service Not starting.

A web application service called 'myapp' failed to start after a server reboot. What commands would you run to diagnose the issue? Write at least 4 commands in order.

1. `systemctl status myapp` (Why: Check if active, failed, or stopped)
2. `journalctl -u myapp -n 50` (Why: Read recent logs to find error messages)
3. `systemctl enabled myapp` (Why: Verify if it starts on boot).
4. `systemctl list-units --type=service` (Why: Confirm service exists and is recognized)

Scenario 2: High CPU Usage.

Your manager reports that the application server is slow. You SSH into the server. What commands would you run to identify which process is using high CPU?

1. Use a command that shows live CPU usage: `top`.
2. Look for processes sorted by CPU percentage: `ps aux --sort=-%cpu | head -10`
3. Easier interactive monitoring: `htop`.
4. Note the PID (process ID) of the top process.

Scenario 3: Finding Docker service Logs.

A developer asks: "Where are the logs for the 'docker' service?" The service is managed by `systemd`. What commands would you use?

1. `systemctl status docker` (why: confirm service state)
2. `journalctl -u docker -n 50` (why: view last 50 log lines)
3. `journalctl -u docker -f`; (why: follow logs in real-time)

Scenario 4: Permission Denied Script

A script at `/home/user/backup.sh` is not executed. When you run it: `./backup.sh`

You get: "Permission denied"? What commands would you use to fix this?

1. Check current permissions: `ls -l /home/user/backup.sh` (Look for `-rw-r--r--` notice no 'x' = not executable.)
2. check for execute permission. `chmod +x /home/user/backup.sh`;
3. verify it worked: `ls -l /home/user/backup.sh` (Look for: `-rwxr-xr-x`)
4. Try running it: `./backup.sh`

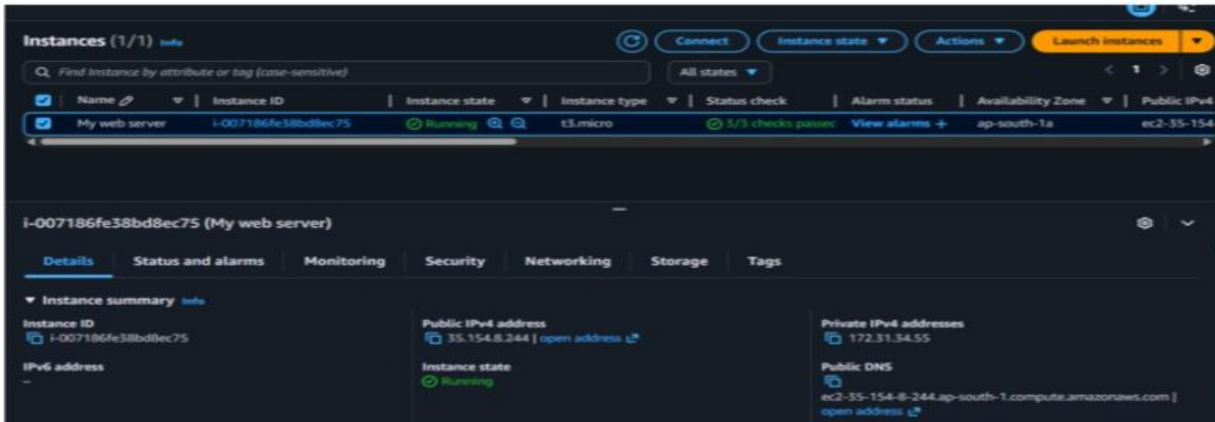
Day 8 – Cloud server setup: Docker, Nginx & web Deployment:

Today's goal is to deploy a real web server on cloud and learn practical server management.

Part 1: Launch cloud instance & SSH Access.

Step 1: Create a cloud instance

Step 2: connect via SSH



Part 2: Install Docker & Nginx

Step 1: update system: `sudo apt update && sudo apt upgrade -y`

Step 2: Install Nginx:

`sudo apt install nginx -y`

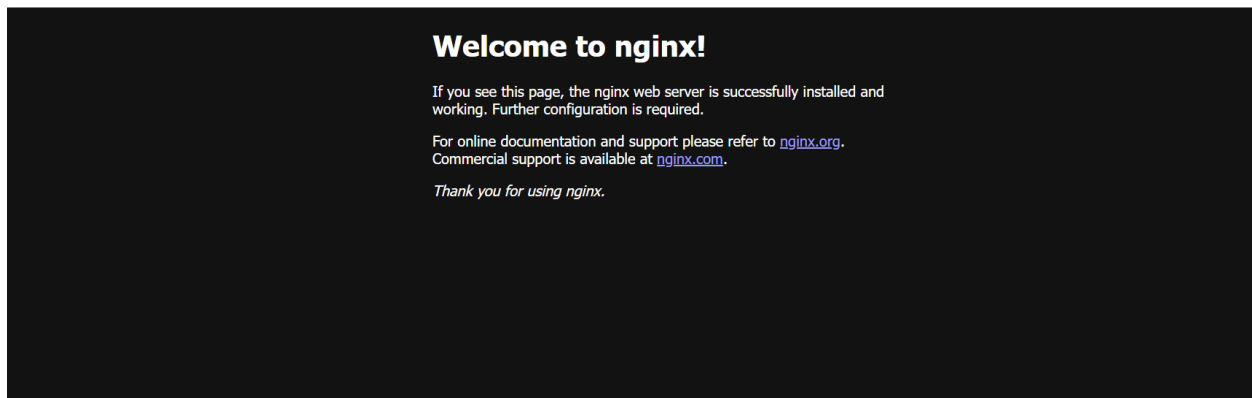
`sudo systemctl start nginx`

`sudo systemctl enable nginx`

Part 3: Security Group Configuration :

Test web Access: open browser and visit – `http://<your-instance-ip>`

You should see the Nginx Welcome Page!



Part 4: Extract Nginx Logs:

Step 1: View Nginx logs.

Step 2 : Save logs to file.

Step 3: Download log file to your local machine.

```
ubuntu@ip-172-31-43-90:~$ sudo tail -f /var/log/nginx/access.log
157.50.144.94 - - [14/Feb/2026:14:36:21 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
157.50.144.94 - - [14/Feb/2026:14:36:22 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://15.206.170.90/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
^C
ubuntu@ip-172-31-43-90:~$ ls /var/log/nginx/
access.log  error.log
ubuntu@ip-172-31-43-90:~$ sudo cat /var/log/nginx/access.log
157.50.144.94 - - [14/Feb/2026:14:36:21 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
157.50.144.94 - - [14/Feb/2026:14:36:22 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://15.206.170.90/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
ubuntu@ip-172-31-43-90:~$ sudo tail -f /var/log/nginx/access.log
157.50.144.94 - - [14/Feb/2026:14:36:21 +0000] "GET / HTTP/1.1" 200 409 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
157.50.144.94 - - [14/Feb/2026:14:36:22 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://15.206.170.90/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
157.50.144.94 - - [14/Feb/2026:14:50:07 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
157.50.144.94 - - [14/Feb/2026:14:50:35 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
157.50.144.94 - - [14/Feb/2026:14:50:40 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
157.50.144.94 - - [14/Feb/2026:14:50:43 +0000] "GET / HTTP/1.1" 304 0 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36"
i-0fb93545bb799043b (My web server)
```

Common Problems :

1. Permission denied (ssh)
2. Cannot access logs.

What I Learned:

1. server management.
2. Firewall/security rules.
3. Log extraction.
4. Production-style troubleshooting

Day 9 – Linux User & Group Management Challenge.

- Create users and set passwords
- Create groups and assign users
- Set up shared directories with group permissions.

Task 1: Create Users:

1. Create three users with home directories and passwords:

tokyo berlin professor

=> sudo useradd -m tokyo sudo useradd -m berlin sudo useradd -m professor -m creates /home/username

Set Passwords: sudo passwd tokyo sudo passwd berlin sudo passwd professor

verify users: cat /etc/passwd | grep -E "tokyo|berlin|professor"

check home directories: ls -l /home

Task 2: Create Groups:

sudo groupadd developers sudo groupadd admins

verify: cat /etc/group | grep -E "developers|admins"

Task 3: Assign users to groups

add users to groups: sudo usermod -aG developers tokyo sudo usermod -aG developers berlin sudo usermod -aG admins berlin sudo usermod -aG admins professor

aG : append to group

Task 4: Shared Directory.

1.Create Directory - sudo mkdir /opt/dev-project

2.Change Group Owner: sudo chgrp developers /opt/dev-project

3.Set permissions: sudo chmod 775 /opt/dev-project. verify: ls -ld /opt/dev-project

4.Test File Creation: Test as tokoyo : sudo -u tokyo touch /opt/dev-project/tokyo-file.txt
Test as berlin: sudo -u berlin touch /opt/dev-project/berlin-file.txt verify: ls -l /opt/dev-project

Task 5: Team Workspace

Create User: sudo useradd -m nairobi sudo passwd nairobi

Create Group: sudo groupadd project-team

Add users to Group: `sudo usermod -aG project-team nairobi` `sudo usermod -aG project-team tokyo`

verify: `groups nairobi` `groups tokyo`

4.Create Workspace Directory: `sudo mkdir /opt/team-workspace` `sudo chgrp project-team /opt/team-workspace` `sudo chmod 775 /opt/team-workspace`

5.Test as nairobi : `sudo -u nairobi touch /opt/team-workspace/test.txt` verify: `ls -l /opt/team-workspace`

```
sri@LAPTOP-KH3CPCF0: ~  
sri@LAPTOP-KH3CPCF0:~$ sudo useradd -m tokyo  
sudo useradd -m berlin  
sudo useradd -m professor  
[sudo] password for sri:  
sri@LAPTOP-KH3CPCF0:~$ sudo passwd tokyo  
sudo passwd berlin  
sudo passwd professor  
New password:  
Retype new password:  
passwd: password updated successfully  
New password:  
Retype new password:  
passwd: password updated successfully  
New password:  
Retype new password:  
passwd: password updated successfully  
sri@LAPTOP-KH3CPCF0:~$ cat /etc/passwd | grep -E "tokyo|berlin|professor"  
tokyo:x:1005:1006:./home/tokyo:/bin/sh  
berlin:x:1006:1007:./home/berlin:/bin/sh  
professor:x:1007:1008:./home/professor:/bin/sh  
sri@LAPTOP-KH3CPCF0:~$ ls -l /home  
total 32  
drwxr-x--- 2 berlin berlin 4096 Feb 15 18:32 berlin  
drwxr-x--- 2 professor professor 4096 Feb 15 18:32 professor  
drwxr-x--- 31 sri sri 4096 Feb 14 15:44 sri  
drwxr-x--- 2 tokoyo tokoyo 4096 Feb 12 18:50 tokoyo  
drwxr-x--- 2 tokyo tokyo 4096 Feb 15 18:32 tokyo  
drwxr-x--- 2 user1 user1 4096 Feb 15 10:17 user1  
drwxr-x--- 2 user2 user2 4096 Feb 15 10:18 user2  
drwxr-x--- 2 virat virat 4096 Feb 12 18:42 virat  
sri@LAPTOP-KH3CPCF0:~$ sudo groupadd developers  
sudo groupadd admins  
[sudo] password for sri:  
sri@LAPTOP-KH3CPCF0:~$ cat /etc/group | grep -E "developers|admins"  
developers:x:1009:  
admins:x:1010:  
sri@LAPTOP-KH3CPCF0:~$ sudo usermod -aG developers tokyo  
sudo usermod -aG developers berlin  
sudo usermod -aG admins berlin  
sudo usermod -aG admins professor  
sri@LAPTOP-KH3CPCF0:~$ groups tokyo  
groups berlin  
groups professor  
tokyo : tokoyo developers  
berlin : berlin developers admins
```

```

sri@LAPTOP-KH3CPCFO:~$ sudo usermod -aG developers tokyo
sudo usermod -aG developers berlin
sudo usermod -aG admins berlin
sudo usermod -aG admins professor
sri@LAPTOP-KH3CPCFO:~$ groups tokyo
groups berlin
groups professor
tokyo : tokyo developers
berlin : berlin developers admins
professor : professor admins
sri@LAPTOP-KH3CPCFO:~$ sudo mkdir /opt/dev-project
sri@LAPTOP-KH3CPCFO:~$ sudo chgrp developers /opt/dev-project
sri@LAPTOP-KH3CPCFO:~$ sudo chmod 775 /opt/dev-project
sri@LAPTOP-KH3CPCFO:~$ ls -ld /opt/dev-project
drwxrwxr-x 2 root developers 4096 Feb 15 18:52 /opt/dev-project
sri@LAPTOP-KH3CPCFO:~$ sudo -u tokyo touch /opt/dev-project/tokyo-file.txt
sri@LAPTOP-KH3CPCFO:~$ sudo -u berlin touch /opt/dev-project/berlin-file.txt
sri@LAPTOP-KH3CPCFO:~$ ls -l /opt/dev-project
total 0
-rw-rw-r-- 1 berlin berlin 0 Feb 15 18:53 berlin-file.txt
-rw-rw-r-- 1 tokyo tokyo 0 Feb 15 18:53 tokyo-file.txt
sri@LAPTOP-KH3CPCFO:~$ sudo useradd -m nairobi
sudo passwd nairobi
New password:
Retype new password:
passwd: password updated successfully
sri@LAPTOP-KH3CPCFO:~$ sudo groupadd project-team
sri@LAPTOP-KH3CPCFO:~$ sudo usermod -aG project-team nairobi
sudo usermod -aG project-team tokyo
sri@LAPTOP-KH3CPCFO:~$ groups nairobi
groups tokyo
nairobi : nairobi project-team
tokyo : tokyo developers project-team
sri@LAPTOP-KH3CPCFO:~$ sudo mkdir /opt/team-workspace
sudo chgrp project-team /opt/team-workspace
sudo chmod 775 /opt/team-workspace
sri@LAPTOP-KH3CPCFO:~$ sudo -u nairobi touch /opt/team-workspace/test.txt
sri@LAPTOP-KH3CPCFO:~$ ls -l /opt/team-workspace
total 0
-rw-rw-r-- 1 nairobi nairobi 0 Feb 15 18:54 test.txt
sri@LAPTOP-KH3CPCFO:~$

```

Day 10 – File Permissions & File Operations Challenge:

Task 1: Create Files

1. Create empty file devops.txt using touch command: touch devops.txt
2. Create notes.txt with some content using cat or echo : echo "Linux permissions practice" > notes.txt
3. Create script.sh using vim with content: echo "Hello DevOps"
4. Verify: ls -l to see permissions.

```
sri@LAPTOP-KH3CPCFO: ~  
sri@LAPTOP-KH3CPCFO:~$ touch devops.txt  
sri@LAPTOP-KH3CPCFO:~$ touch notes.txt  
sri@LAPTOP-KH3CPCFO:~$ cat notes.txt  
Linux Day 06 Practice  
Learning file operations  
Using redirection operators  
Practicing tee command  
Reading file using cat  
Reading first lines using head  
Reading last lines using tail  
Linux fundamentals are important  
Consistency builds mastery  
sri@LAPTOP-KH3CPCFO:~$ vim notes.txt  
sri@LAPTOP-KH3CPCFO:~$ script.sh  
script.sh: command not found  
sri@LAPTOP-KH3CPCFO:~$ vim script.sh  
sri@LAPTOP-KH3CPCFO:~$ ls -l  
total 56756  
drwxr-xr-x  5 sri sri      4096 Jul  7  2025 AWS-Session  
drwxr-xr-x  3 sri sri      4096 Jan 26 18:17 Ansible  
-rw-r--r--  1 root root    2879 Jan 31 18:00 Sri  
-rw-r--r--  1 sri sri     1720 Aug 26  2024 Systeminfo  
drwxr-xr-x  3 sri sri      4096 Aug 26  2024 Test  
drwxr-xr-x  2 sri sri      4096 Jan 26 17:03 ansible-first  
-rw-r--r--  1 sri sri        0 Feb 18  2025 bash  
-rw-r--r--  1 sri sri        0 Feb 18  2025 cd  
-rw-r--r--  1 sri sri        0 Feb 16 19:44 devops.txt  
drwxr-xr-x  3 sri sri      4096 Aug 26  2024 git-tutorial  
drwxr-xr-x  2 sri sri      4096 Aug 26  2024 hello  
-rw-r--r--  1 sri sri        0 Sep  1  2024 hi  
drwxr-xr-x  2 sri sri      4096 Jan 26 19:00 keys  
-rw-r--r--  1 sri sri     86016 Sep  3  2024 kubectl  
-rw-r--r--  1 sri sri    57886208 Sep  3  2024 kubectl.exe  
-rw-r--r--  1 sri sri        64 Sep  3  2024 kubectl.exe.sha256  
drwxr-xr-x 10 sri sri      4096 Jan 27 15:51 kubernetes  
-rw-r--r--  1 sri sri        29 Feb 11 09:50 last_run.txt  
drwxr-xr-x  2 sri sri      4096 Sep  1  2024 my-docker-setup  
drwxr-xr-x  2 sri sri      4096 Sep  1  2024 my_docker_project
```

Task 2: Read Files.

1. Read notes.txt using cat: cat notes.txt
2. View script.sh in vim read-only mode: vim -R script.sh
3. Display first 5 lines of /etc/passwd: head -n 5 /etc/passwd
4. Display last 5 lines:tail -n 5 /etc/passwd.

```
sri@LAPTOP-KH3CPCFO: ~  
drwxr-xr-x  3 sri sri      4096 Sep  1  2024 wordpress-docker  
sri@LAPTOP-KH3CPCFO:~$ vim -R script.sh  
sri@LAPTOP-KH3CPCFO:~$ head -n 5 /etc/passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
sri@LAPTOP-KH3CPCFO:~$ tail -n 5 /etc/passwd  
user2:x:1004:1005:jaanu,1,1,1:/home/user2:/bin/bash  
tokyo:x:1005:1006:./home/tokyo:/bin/sh  
berlin:x:1006:1007:./home/berlin:/bin/sh  
professor:x:1007:1008:./home/professor:/bin/sh  
nairobi:x:1008:1011:./home/nairobi:/bin/sh  
sri@LAPTOP-KH3CPCFO:~$ ls -l devops.txt notes.txt script.sh  
-rw-r--r--  1 sri sri    0 Feb 16 19:44 devops.txt  
-rw-r--r--  1 sri sri  242 Feb 16 19:45 notes.txt  
-rw-r--r--  1 sri sri   21 Feb 16 19:46 script.sh
```

Task 3: Understand Permissions

Format: rwxrwxrwx(owner-group-others)

R = read (4), W=write (2), X=execute (1)

Check your files: `ls -l devops.txt notes.txt script.sh`

Task 4: Modify Permissions

1. Make script.sh executable: `chmod +x script.sh` and verify `ls -l script.sh`
2. Make devops.txt read-only: `chmod a-w devops.txt` and verify `ls -l devops.txt`
3. Set notes.txt to 640: `chmod 640 notes.txt` and verify: `ls -l notes.txt`
4. Create project directory with 755: `mkdir project, chmod 755 project.`

Task 5: Test Permissions

1. Try Writing to Read-Only File: `echo "test" >> devops.txt`
2. Try Executing Without Execute Permission:
Remove execute: `chmod -x script.sh`
Try running: `./script.sh`
Output: Permission denied.
Reason: Execute bit is required to run scripts.

```
sri@LAPTOP-KH3CPCFO: ~  
-rw-r--r-- 1 sri sri 21 Feb 16 19:46 script.sh  
sri@LAPTOP-KH3CPCFO:~$ chmod +x script.sh  
sri@LAPTOP-KH3CPCFO:~$ ls -l script.sh  
-rwxr-xr-x 1 sri sri 21 Feb 16 19:46 script.sh  
sri@LAPTOP-KH3CPCFO:~$ ./script.sh  
hello DevOps  
sri@LAPTOP-KH3CPCFO:~$ chmod a-w devops.txt  
sri@LAPTOP-KH3CPCFO:~$ ls -l devops.txt  
-r--r--r-- 1 sri sri 0 Feb 16 19:44 devops.txt  
sri@LAPTOP-KH3CPCFO:~$ chmod 640 notes.txt  
sri@LAPTOP-KH3CPCFO:~$ ls -l notes.txt  
-rw-r----- 1 sri sri 242 Feb 16 19:45 notes.txt  
sri@LAPTOP-KH3CPCFO:~$ mkdir project  
sri@LAPTOP-KH3CPCFO:~$ chmod 755 project  
sri@LAPTOP-KH3CPCFO:~$ ls -ld project  
drwxr-xr-x 2 sri sri 4096 Feb 16 20:11 project  
sri@LAPTOP-KH3CPCFO:~$ echo "test" >> devops.txt  
-bash: devops.txt: Permission denied  
sri@LAPTOP-KH3CPCFO:~$ chmod -x script.sh  
sri@LAPTOP-KH3CPCFO:~$ ./script.sh  
-bash: ./script.sh: Permission denied  
sri@LAPTOP-KH3CPCFO:~$ ls -l  
total 56760  
drwxr-xr-x 5 sri sri 4096 Jul 7 2025 AWS-Session  
drwxr-xr-x 3 sri sri 4096 Jan 26 18:17 Ansible  
-rw-r--r-- 1 root root 2879 Jan 31 18:00 Sri  
-rw-r--r-- 1 sri sri 1720 Aug 26 2024 Systeminfo  
drwxr-xr-x 3 sri sri 4096 Aug 26 2024 Test  
drwxr-xr-x 2 sri sri 4096 Jan 26 17:03 ansible-first  
-rw-r--r-- 1 sri sri 0 Feb 18 2025 bash  
-rw-r--r-- 1 sri sri 0 Feb 18 2025 cd  
-r--r--r-- 1 sri sri 0 Feb 16 19:44 devops.txt  
drwxr-xr-x 3 sri sri 4096 Aug 26 2024 git-tutorial  
drwxr-xr-x 2 sri sri 4096 Aug 26 2024 hello
```

Day 11 – File ownership challenge (chown & chgrp)

Task 1: Understanding Ownership

1. Run `ls -l` in your home directory.
2. Identify the owner and group columns.
3. Check who owns your files.

Format: `-rw-r- -r- -1 owner group size date filename.`

Difference Between Owner and Group:

Owner – The user who owns the file. Has primary control.

Group – A collection of users. Members inherit group-level permissions.

Task 2: Basic chown Operations

1. Create file: `touch devops-file.txt`

2. Check owner: `ls -l devops-file.txt => -rw-r--r-- 1 ubuntu ubuntu devops-file.txt`
3. Change owner to Tokyo: `sudo chown tokyo devops-file.txt`
Verify: `ls -l devops-file.txt`
4. Change owner to berlin : `sudo chown berlin devops-file.txt` (Verify again)

Task 3: Basic chgrp Operations

1. Create file: `touch team-notes.txt`
2. Check group: `ls -l team-notes.txt`
3. Create group: `sudo groupadd heist-team`
4. Change group: `sudo chgrp heist-team team-notes.txt` (Verify: `ls -l team-notes.txt`)
Group column should now show: heist-team

Task 4: Combined Owner & Group Change.

1. Create file: `touch project-config.yaml`
2. Change both owner & group: `sudo chown professor:heist-team project-config.yaml` (Verify it)
3. Create directory: `mkdir app-logs`
4. Change ownership: `sudo chown berlin:heist-team app-logs` (Verify: `ls -ld app-logs`)

Task 5: Recursive Ownership

1. Create directory structure :

`mkdir -p heist-project/vault`

`mkdir -p heist-project/plans`

`touch heist-project/vault/gold.txt`

`touch heist-project/plans/strategy.conf`

2. Create group: `sudo groupadd planners`

3. Change ownership recursively: `sudo chown -R professor:planners heist-project/`

(Verify: `ls -lR heist-project/`)

Task 6: Practice Challenge

1. Create users:

`sudo useradd -m tokyo`

`sudo useradd -m berlin`

`sudo useradd -m nairobi`

2. Create groups: `sudo groupadd vault-team`

`Sudo groupadd tech-team`

3. Create directory: `mkdir bank-heist`

Set ownership:

1. access-code.txt: `sudo chown tokyo:vault-team bank-heist/access-codes.txt`

2. blueprints.pdf: `sudo chown berlin:tech-team bank-heist/blueprints.pdf`

3. escape-plan.txt : `sudo chown nairobi:vault-team bank-heist/escape-plan.txt` Verify : `ls -l bank-heist/`

```
sri@LAPTOP-KH3CPCFO: ~  
sri@LAPTOP-KH3CPCFO:~$ touch devops-file.txt  
sri@LAPTOP-KH3CPCFO:~$ ls -l devops-file.txt  
-rw-r--r-- 1 sri sri 0 Feb 16 20:15 devops-file.txt  
sri@LAPTOP-KH3CPCFO:~$ sudo chown tokoyo devops-file.txt  
[sudo] password for sri:  
sri@LAPTOP-KH3CPCFO:~$ ls -l devops-file.txt  
-rw-r--r-- 1 tokoyo sri 0 Feb 16 20:15 devops-file.txt  
sri@LAPTOP-KH3CPCFO:~$ sudo chown berlin devops-file.txt  
sri@LAPTOP-KH3CPCFO:~$ ls -l devops-file.txt  
-rw-r--r-- 1 berlin sri 0 Feb 16 20:15 devops-file.txt  
sri@LAPTOP-KH3CPCFO:~$ touch team-notes.txt  
sri@LAPTOP-KH3CPCFO:~$ ls -l team-notes.txt  
-rw-r--r-- 1 sri sri 0 Feb 16 20:20 team-notes.txt  
sri@LAPTOP-KH3CPCFO:~$ sudo groupadd heist-team  
sri@LAPTOP-KH3CPCFO:~$ sudo chgrp heist-team team-notes.txt  
sri@LAPTOP-KH3CPCFO:~$ sudo chgrp heist-team team-notes.txt  
sri@LAPTOP-KH3CPCFO:~$ ls -l team-notes.txt  
-rw-r--r-- 1 sri heist-team 0 Feb 16 20:20 team-notes.txt  
sri@LAPTOP-KH3CPCFO:~$ touch project-config.yaml  
sri@LAPTOP-KH3CPCFO:~$ sudo chown professor:heist-team project-config.yaml  
sri@LAPTOP-KH3CPCFO:~$ ls -l project-config.yaml  
-rw-r--r-- 1 professor heist-team 0 Feb 16 20:23 project-config.yaml  
sri@LAPTOP-KH3CPCFO:~$ mkdir app-logs  
sri@LAPTOP-KH3CPCFO:~$ sudo chown berlin:heist-team app-logs  
sri@LAPTOP-KH3CPCFO:~$ ls -ld app-logs  
drwxr-xr-x 2 berlin heist-team 4096 Feb 16 20:24 app-logs  
sri@LAPTOP-KH3CPCFO:~$ mkdir -p heist-project/vault  
mkdir -p heist-project/plans  
touch heist-project/vault/gold.txt  
touch heist-project/plans/strategy.conf  
sri@LAPTOP-KH3CPCFO:~$ sudo groupadd planners  
sri@LAPTOP-KH3CPCFO:~$ sudo chown -R professor:planners heist-project/  
sri@LAPTOP-KH3CPCFO:~$ ls -lR heist-project/  
heist-project/  
total 8  
drwxr-xr-x 2 professor planners 4096 Feb 16 20:25 plans  
drwxr-xr-x 2 professor planners 4096 Feb 16 20:25 vault  
  
heist-project/plans:  
total 0  
-rw-r--r-- 1 professor planners 0 Feb 16 20:25 strategy.conf  
  
sri@LAPTOP-KH3CPCFO:~$ sudo groupadd planners  
sri@LAPTOP-KH3CPCFO:~$ sudo chown -R professor:planners heist-project/  
sri@LAPTOP-KH3CPCFO:~$ ls -lR heist-project/  
heist-project/  
total 8  
drwxr-xr-x 2 professor planners 4096 Feb 16 20:25 plans  
drwxr-xr-x 2 professor planners 4096 Feb 16 20:25 vault  
  
heist-project/plans:  
total 0  
-rw-r--r-- 1 professor planners 0 Feb 16 20:25 strategy.conf  
  
heist-project/vault:  
total 0  
-rw-r--r-- 1 professor planners 0 Feb 16 20:25 gold.txt  
sri@LAPTOP-KH3CPCFO:~$ sudo useradd -m tokoyo  
sudo useradd -m berlin  
sudo useradd -m nairobi  
useradd: user 'tokyo' already exists  
useradd: user 'berlin' already exists  
useradd: user 'nairobi' already exists  
sri@LAPTOP-KH3CPCFO:~$ sudo groupadd vault-team  
sudo groupadd tech-team  
sri@LAPTOP-KH3CPCFO:~$ mkdir bank-heist  
sri@LAPTOP-KH3CPCFO:~$ touch bank-heist/access-codes.txt  
touch bank-heist/blueprints.pdf  
touch bank-heist/escape-plan.txt  
sri@LAPTOP-KH3CPCFO:~$ sudo chown tokoyo:vault-team bank-heist/access-codes.t  
xt  
sri@LAPTOP-KH3CPCFO:~$ sudo chown berlin:tech-team bank-heist/blueprints.pdf  
sri@LAPTOP-KH3CPCFO:~$ sudo chown nairobi:vault-team bank-heist/escape-plan.  
txt  
sri@LAPTOP-KH3CPCFO:~$ ls -l bank-heist/  
total 0  
-rw-r--r-- 1 tokoyo vault-team 0 Feb 16 20:26 access-codes.txt  
-rw-r--r-- 1 berlin tech-team 0 Feb 16 20:26 blueprints.pdf  
-rw-r--r-- 1 nairobi vault-team 0 Feb 16 20:26 escape-plan.txt  
sri@LAPTOP-KH3CPCFO:~$ |
```

Day 12 – Breather & Revision:

1. Mindset & Plan Review
2. Processes & Services Review:

1. Check running processes: `ps aux | head`
2. Check service status: `systemctl status ssh`
3. Check service logs: `journalctl -u ssh --since "10 minutes ago"`
3. File Skills Quick Practice:
 1. Append content: `echo "Revision entry" >> notes.txt`
 2. Change permission: `chmod 640 notes.txt`
 3. Change ownership: `sudo chown tokyo:vault-team notes.txt` (verify it)

4. Cheat Sheet Refresh – Top 5 Incident Commands

1. `ps aux` , 2. `systemctl status <service>` , 3. `journalctl -xe` , 4. `df -h` , 5. `ls -l`

5. User / Group Sanity Practice :

Scenario Recreated: `sudo useradd -m testuser` , `sudo groupadd testgroup` , `sudo chown testuser:testgroup notes.txt`

Verification: `id testuser` , `ls -l notes.txt`

Mini Self-Check :

1. Which 3 commands save you the most time?
`Systemctl status` , `ls -l` , `ps aux` => These diagnose 80% of issues quickly.
2. How do you check if a service is healthy?
`systemctl status <service>`
`journalctl -u <service> -n 20`
`ss -tuln | grep <port>`
3. How do you safely change ownership & permissions?
`sudo chown user:group file.txt`
`chmod 640 file.txt`

Day –13: Linux Volume Management (LVM)

Create a EC2 instance and add 3 volumes in EBS(10g,12g,14g) attach the volumes to instance and give space name (/dev/sdf , /dev/sdg , /dev/sdh)

Task 1- Check current storage Run: lsblk, sudo pvs, sudo vgs, sudo lvs , df -h

```
ubuntu@ip-172-31-26-129: ~  
ubuntu@ip-172-31-26-129:~$ lsblk  
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS  
loop0                 7:0      0    74M  1 loop /snap/core22/2163  
loop1                 7:1      0   27.6M  1 loop /snap/amazon-ssm-agent/11797  
loop2                 7:2      0   50.9M  1 loop /snap/snapd/25577  
nvme0n1              259:0     0     8G   0 disk  
├─nvme0n1p1          259:1     0     7G   0 part /  
├─nvme0n1p14         259:2     0     4M   0 part  
├─nvme0n1p15         259:3     0   106M   0 part /boot/efi  
└─nvme0n1p16         259:4     0   913M   0 part /boot  
ubuntu@ip-172-31-26-129:~$ sudo pvs  
ubuntu@ip-172-31-26-129:~$ sudo vgs  
ubuntu@ip-172-31-26-129:~$ sudo lvs  
ubuntu@ip-172-31-26-129:~$ df -hh  
Filesystem           Size  Used Avail Use% Mounted on  
/dev/root             6.8G  1.8G  5.0G  27% /  
tmpfs                 458M      0  458M   0% /dev/shm  
tmpfs                 183M  876K  182M   1% /run  
tmpfs                 5.0M      0   5.0M   0% /run/lock  
efivarfs             128K   3.6K  120K   3% /sys/firmware/efi/efivars  
/dev/nvme0n1p16       881M   89M   730M  11% /boot  
/dev/nvme0n1p15       105M   6.2M   99M   6% /boot/efi  
tmpfs                 92M    12K   92M   1% /run/user/1000  
ubuntu@ip-172-31-26-129:~$
```

Task 2 – Create Physical volume

```
ubuntu@ip-172-31-1-175:~$ lsblk  
NAME                MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS  
loop0                 7:0      0   27.6M  1 loop /snap/amazon-ssm-agent/11797  
loop1                 7:1      0    74M  1 loop /snap/core22/2163  
loop2                 7:2      0   50.9M  1 loop /snap/snapd/25577  
nvme0n1              259:0     0     8G   0 disk  
├─nvme0n1p1          259:1     0     7G   0 part /  
├─nvme0n1p14         259:2     0     4M   0 part  
├─nvme0n1p15         259:3     0   106M   0 part /boot/efi  
└─nvme0n1p16         259:4     0   913M   0 part /boot  
nvme1n1              259:5     0    10G   0 disk  
nvme2n1              259:6     0    12G   0 disk  
nvme3n1              259:7     0    14G   0 disk  
ubuntu@ip-172-31-1-175:~$
```

Whenever you need to Run LVM do sudo su so you run in your terminal.

Now Run pvs to see physical volume display but there is no one.

pvcreate /dev/nvme1n1 /dev/nvme2n1 /dev/nvme3n1 (physical volumes created).

Pvs (displays physical vouldmes)

```

EXTING.
ubuntu@ip-172-31-1-175:~$ sudo su
root@ip-172-31-1-175:/home/ubuntu# lvm
lvm> pvs
lvm> pvcreate /dev/nvme1n1 /dev/nvme2n1 /dev/nvme3n1
Physical volume "/dev/nvme1n1" successfully created.
Physical volume "/dev/nvme2n1" successfully created.
Physical volume "/dev/nvme3n1" successfully created.
lvm> |

```

Task 3: Create Volume Group:

To create volume group run : `vgcreate tws_vg (name of the group ) /dev/nvme1n1 /dev/nvme2n1` (combining these two volumes to create volume group)

```

root@ip-172-31-1-175:/home/ubuntu# lvm
lvm> vgcreate tws_vg /dev/nvme1n1 /dev/nvme2n1
Volume group "tws_vg" successfully created
lvm> vgs
VG      #PV #LV #SN Attr   VSize  VFree
tws_vg   2   0   0 wz--n- 21.99g 21.99g
lvm> |

```

Task 4: Create Logical volume Group:

Run the command: `lvcreate -L 500MB -n tws_lv tws_vg`

```
lvm> lvcreate -L 500MB -n tws_lv tws_vg
Logical volume "tws_lv" created.
lvm> |
```

```
root@ip-172-31-1-175:/home/ubuntu# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                 7:0    0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1                 7:1    0   74M  1 loop /snap/core22/2163
loop2                 7:2    0 50.9M  1 loop /snap/snapd/25577
nvme0n1              259:0    0    8G  0 disk
├─nvme0n1p1          259:1    0    7G  0 part /
├─nvme0n1p14         259:2    0    4M  0 part
├─nvme0n1p15         259:3    0 106M  0 part /boot/efi
└─nvme0n1p16         259:4    0 913M  0 part /boot
nvme1n1              259:5    0   10G  0 disk
└─tws_vg-tws_lv      252:0    0 500M  0 lvm
nvme2n1              259:6    0   12G  0 disk
nvme3n1              259:7    0   14G  0 disk
root@ip-172-31-1-175:/home/ubuntu# lvs
  LV      VG      Attr      LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
  tws_lv  tws_vg  -wi-a----- 500.00m
```

Task 5: Format and Mount

Create an empty directory for mounting purposes. (`mkdir /mnt/tws_lv_mount`)

Now let's format lv (logical volume)

=> `mkfs.ext4 /dev/tws_vg/tws_lv` (`mkfs` = make file system , `ext4` = filesystem type (fourth extended filesystem))

Now let's mount mount source to destination .

=> `mount /dev/tws_vg/tws_lv /mnt/tws_lv_mount`

```

root@ip-172-31-1-175: /home/ubuntu
root@ip-172-31-1-175:/home/ubuntu# sudo mkdir -p /mnt/tws_lv_mount
root@ip-172-31-1-175:/home/ubuntu# sudo mount /dev/tws_vg/tws_lv /mnt/tws_lv_mount
root@ip-172-31-1-175:/home/ubuntu# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                              7:0      0 27.6M 1 loop /snap/amazon-ssm-agent/11797
loop1                              7:1      0  74M 1 loop /snap/core22/2163
loop2                              7:2      0 50.9M 1 loop /snap/snapd/25577
nvme0n1                            259:0    0    8G 0 disk
├─nvme0n1p1                        259:1    0    7G 0 part /
├─nvme0n1p14                       259:2    0    4M 0 part
├─nvme0n1p15                       259:3    0  106M 0 part /boot/efi
└─nvme0n1p16                       259:4    0  913M 0 part /boot
nvme1n1                            259:5    0   10G 0 disk
└─tws_vg-tws_lv                   252:0    0  500M 0 lvm  /mnt/tws_lv_mount
nvme2n1                            259:6    0   12G 0 disk
nvme3n1                            259:7    0   14G 0 disk
root@ip-172-31-1-175:/home/ubuntu# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  1.8G  5.0G  27% /
tmpfs            458M    0  458M   0% /dev/shm
tmpfs            183M  932K  182M   1% /run
tmpfs            5.0M    0   5.0M   0% /run/lock
efivarfs         128K  3.6K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M   89M   730M  11% /boot
/dev/nvme0n1p15  105M   6.2M   99M   6% /boot/efi
tmpfs            92M   12K   92M   1% /run/user/1000
/dev/mapper/tws_vg-tws_lv 452M  24K  417M   1% /mnt/tws_lv_mount
root@ip-172-31-1-175:/home/ubuntu#

```

Task 6: Extend the volume:

Add an extra 200 MB to another volume.

Lvextend -L +200M /dev/tws_vg/tws_lv

See the output using this command : df -h or lsblk

```

root@ip-172-31-1-175:/mnt# cd
root@ip-172-31-1-175:~# lvm
lvm> lvextend -L +5G /dev/tws_vg/tws_lv
Size of logical volume tws_vg/tws_lv changed from 500.00 MiB (125 extents)
to 5.49 GiB (1405 extents).
Logical volume tws_vg/tws_lv successfully resized.
lvm> lvextend -L +200MB /dev/tws_vg/tws_lv
Size of logical volume tws_vg/tws_lv changed from <5.49 GiB (1405 extents)
to 5.69 GiB (1455 extents).
Logical volume tws_vg/tws_lv successfully resized.
lvm> exit
Exiting.
root@ip-172-31-1-175:~# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                              7:0      0 27.6M 1 loop /snap/amazon-ssm-agent/11797
loop1                              7:1      0  74M 1 loop /snap/core22/2163
loop2                              7:2      0 50.9M 1 loop /snap/snapd/25577
nvme0n1                            259:0    0    8G 0 disk
├─nvme0n1p1                        259:1    0    7G 0 part /
├─nvme0n1p14                       259:2    0    4M 0 part
├─nvme0n1p15                       259:3    0  106M 0 part /boot/efi
└─nvme0n1p16                       259:4    0  913M 0 part /boot
nvme1n1                            259:5    0   10G 0 disk
└─tws_vg-tws_lv                   252:0    0  5.7G 0 lvm  /mnt/tws_lv_mount
nvme2n1                            259:6    0   12G 0 disk
nvme3n1                            259:7    0   14G 0 disk

```

